

KAUSHIK KUMAR

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EDUCATION

UNIVERSITY OF ARIZONA

Master of Science in Data Science, GPA: 4.0

Aug 2024 - Dec 2025

Tucson, AZ

VELS INSTITUTE OF SCIENCE & TECHNOLOGY

Bachelor of Engineering Computer Science and Engineering,

Jun 2019 - May 2023

Chennai, TN

EXPERIENCE

DATA ANALYST

Aug 2025 - Present

ELLER PARTNERSHIPS OFFICE (Client: Banner Health)

Tucson, AZ

- Built 12+ PowerBI dashboards integrating CMS Medicare and ACS SDOH data, enabling directors to track chronic disease prevalence, inpatient demand, and social risk factors across AZ, CO, and UT, and accelerating market-entry evaluations by 40%.
- Developed a composite SDOH risk index (housing, poverty, disability, minority status) using SQL and Python (pandas, statsmodels) ran OLS regression ($R^2=0.93$, $p < 0.01$) to identify housing stress as top predictor of chronic disease across 50+ counties.

DATA PIPELINES SOFTWARE ENGINEER INTERN

Jun 2025 - Aug 2025

SELECTOR AI

Santa Clara, CA

- Engineered end-to-end ingestion pipelines (rsyslog, Promtail, Loki, Kafka, cloud networks) with real-time normalization and NER-based clustering, Pytest validations(50+ tests) ensuring reliable event detection across 1000K+ logs/day. Deployed containerized services to a private GCP registry and delivered to clients via Selector's S2AP platform.
- Migrated Logminer log classification from a Random Forest model to a BM25S retrieval engine (Lucene/Numba), achieving 99.2% accuracy, with inference throughput 6K logs/sec, and a reduced memory footprint of < 1GB.

DATA SCIENTIST

Jun 2023 - Jul 2024

JOHNSON ELECTRIC

Chennai, India

- Built SQL-driven pipelines and Power BI dashboards to monitor End-of-Line (EOL) tests, applying statistical divergence methods (JSD, KL) for anomaly detection with zero false positives, and deploying novel Kalman Filter models for physics-based production line solutions via Dockerized VMs that cut pump testing time by 72% (80W) and 50% (50W), saving 74K\$/month.
- Built Power BI (DirectQuery) dashboards with drilldowns from cells → stations → steps, enabling real-time actuator anomaly detection and reducing manual line stoppages by 60%, saving 10+ hours/week in troubleshooting.

RESEARCH DATA SCIENCE INTERN

Dec 2022 - Feb 2023

NATIONAL UNIVERSITY OF SINGAPORE (Collaboration with HPE)

Singapore

- Analyzed a 200K+ multilingual dataset with n-gram frequency and bias detection metrics, building classification pipelines with mBERT + GRU that achieved 91% detection accuracy while reducing false positives by 28% for underrepresented languages.
- Presented findings to 15+ HPE executives, showcasing AI-powered content moderation; surfaced results via Flask REST APIs, React dashboards and chrome extension on Azure enabling real-time monitoring of 10K+ items/day across multiple languages.

FULLSTACK AND ANALYST INTERN

Jan 2022 - Mar 2022

SUVIDHA FOUNDATION

Chennai, India

- Built a university management dashboard with React (frontend) + PHP (backend), JWT-secured APIs, and Heroku cloud hosting, automating 50+ admin workflows and providing real-time analytics on student performance, faculty reports and institutional data
- Developed Python & SQL scripts to stream time-stamped asset telemetry into SiteConnect app, enabling remote monitoring across 15+ cement plant assets and reducing unplanned downtime by 30% through instantaneous failure alerts.

PROJECTS

UNDERSTANDING DISEASE HETEROGENEITY (Client: C-Path)(GitHub)

Mar 2025 - May 2025

- Performed EDA and multivariate analysis on 50K+ synthetic ALS patient records; applied PCA and autoencoders to reduce feature dimensionality by 82% (120 → 22 features) while preserving 93% variance, isolating latent drivers of disease progression.
- Clustered patients with K-means and hierarchical clustering (silhouette > 0.71), identifying 3 subtypes; built Plotly dashboards to aid clinician interpretability and trial design.

GLOBAL STOCK INDEX PREDICTION AND VOLATILITY ANALYSIS

Aug 2024 - Dec 2024

- Forecasted NASDAQ, NYSE, Nikkei, and NIFTY indices with XGBoost ($R^2 = 0.94$, -18% RMSE vs. ARIMA) and K-means clustering, identifying 4 market segments and visualizing trends through 2030.

TECHNICAL SKILLS

Languages & Cloud: Python, SQL, R, Bash, Git, Javascript, Azure(Devops, ML, Bricks), AWS(S3, Lambda, EC2), DigitalOcean

ETL & BI: MS Excel, Power BI, Tableau, Airflow, dbt, Talend, Apache Kafka, Google BigQuery, Postgres, MongoDB, VectorDB

Data Analytics: Exploratory Data Analysis (EDA), Data Cleaning, Statistical Analysis, A/B Testing, Regression Modeling, Forecasting, Data Modeling, Feature Engineering, Pattern Recognition, KPI Reporting, Matplotlib, Seaborn, PyQt5, Flask, FastAPI, React, REST APIs, JWT, Microsoft Graph API

Certifications: Google - Data Analytics, IBM - Data Analytics Professional